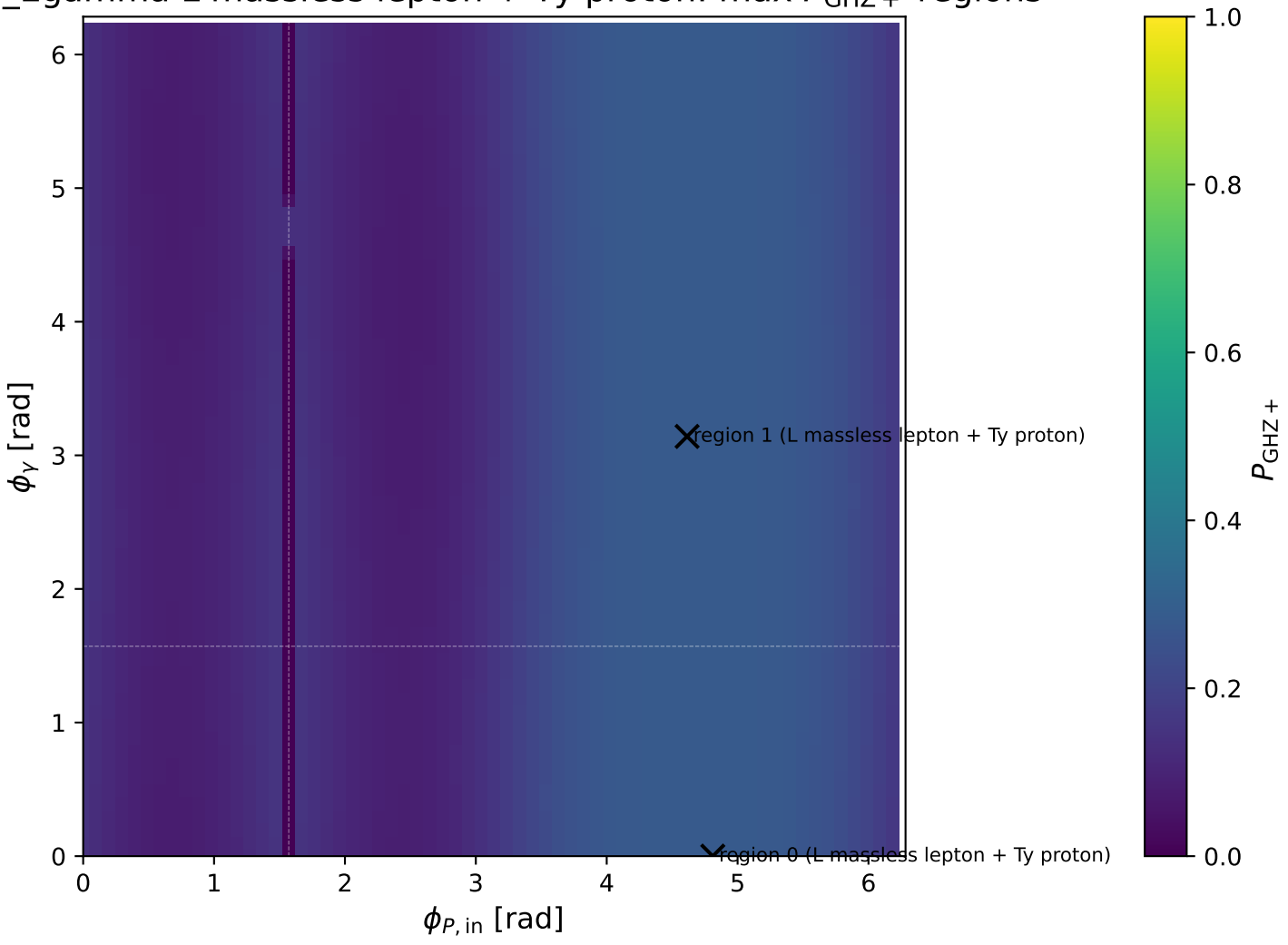
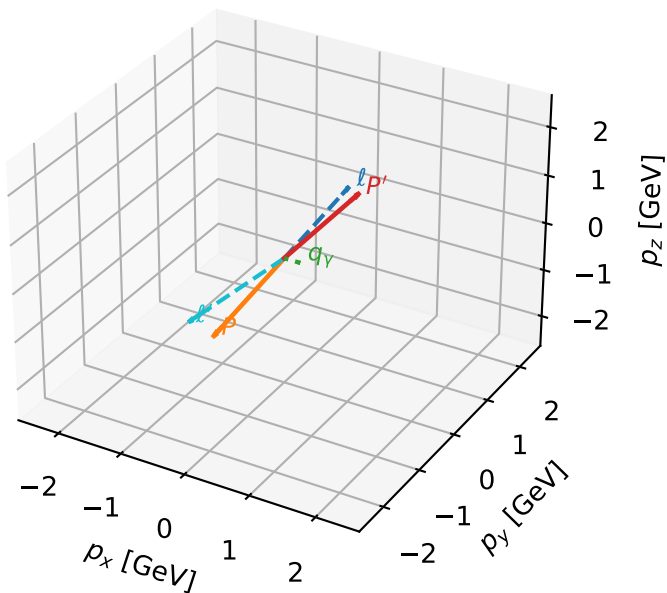


γ L massless lepton + Ty proton: max P_{GHZ} regions



L massless lepton + Ty proton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

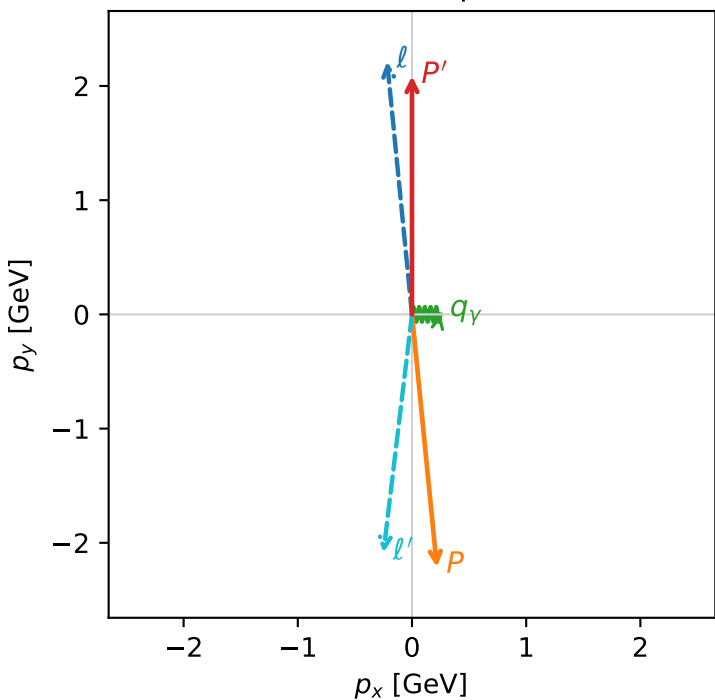


ghz_purity_low_egamma_lty_region_0 $P_{\{\text{rm GHZ+}\}}=0.27977$
region: low_Egamma_LTy_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_{l_0}\rangle \otimes |T_{y,p}\rangle$

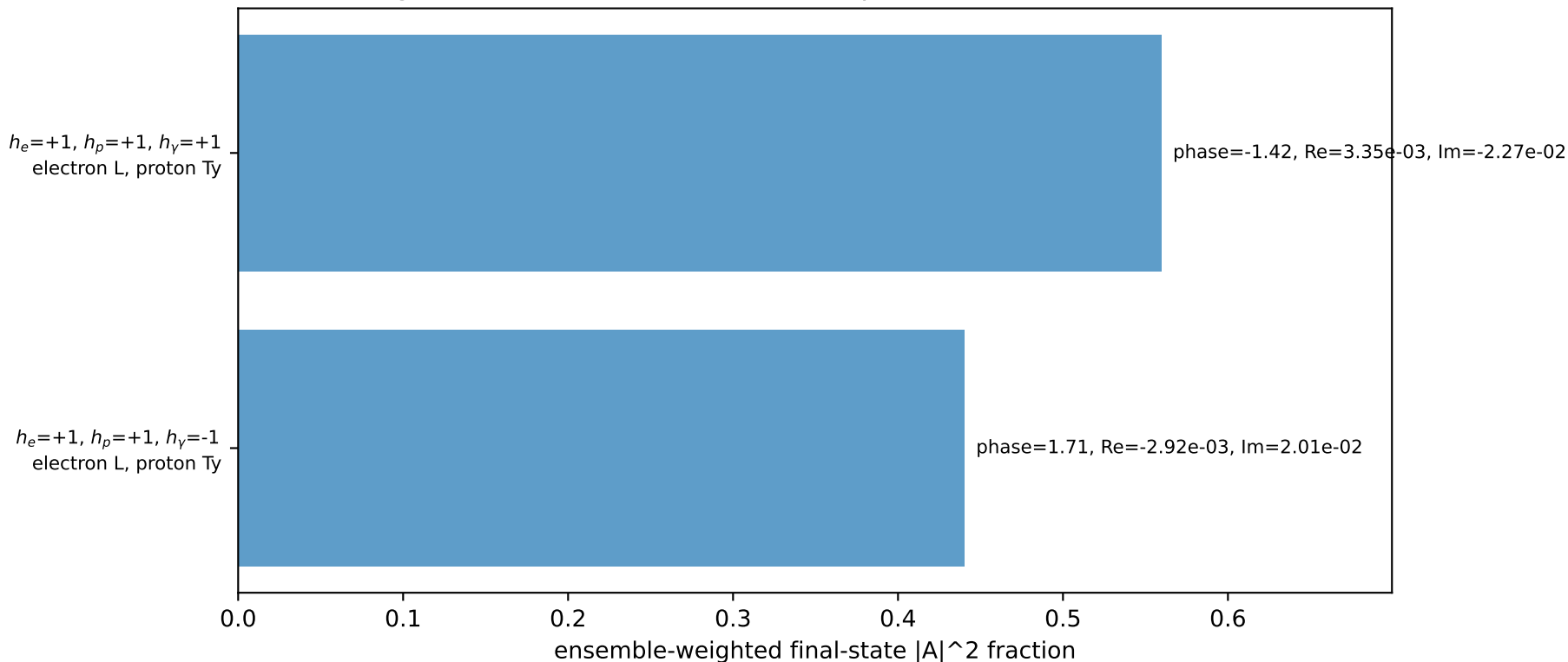
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.08621$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.66897$, $\phi_P=4.81056$
 $E_\gamma=0.25$, $\phi_\gamma=0$
 $Q^2=18.4751$, $x_B=0.941704$, $t=-18.5208$, $W^2=2.02353$, $y=0.950707$
 $\theta(l', \gamma)=96.8334$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= -0.2180$ $p_y= 2.2137$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 0.2180$ $p_y= -2.2137$ $p_z= 0.0000$
 l' $E= 2.1011$ $p_x= -0.2500$ $p_y= -2.0862$ $p_z= 0.0000$
 P' $E= 2.2874$ $p_x= 0.0000$ $p_y= 2.0862$ $p_z= 0.0000$
 q_γ $E= 0.2500$ $p_x= 0.2500$ $p_y= 0.0000$ $p_z= 0.0000$

Transverse plane

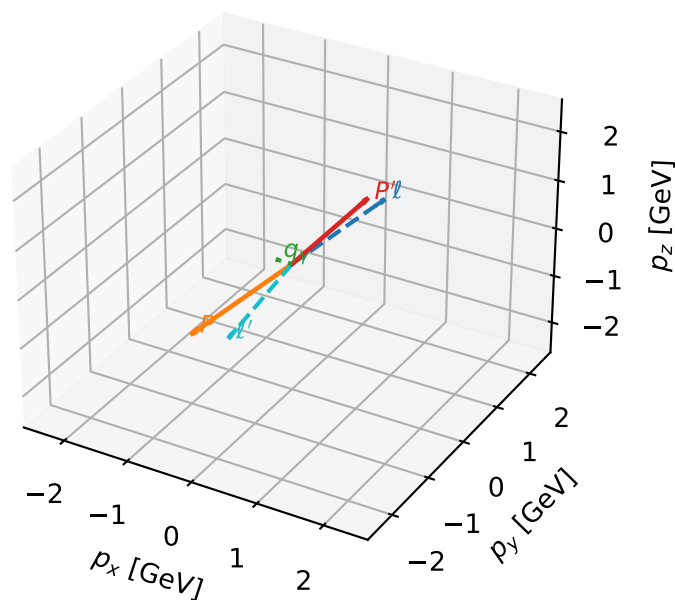


Leading initial-ensemble/final-state components (N=2, retained=100.0%)



L massless lepton + Ty proton characteristic max P_{GHZ} + configuration

3D momenta

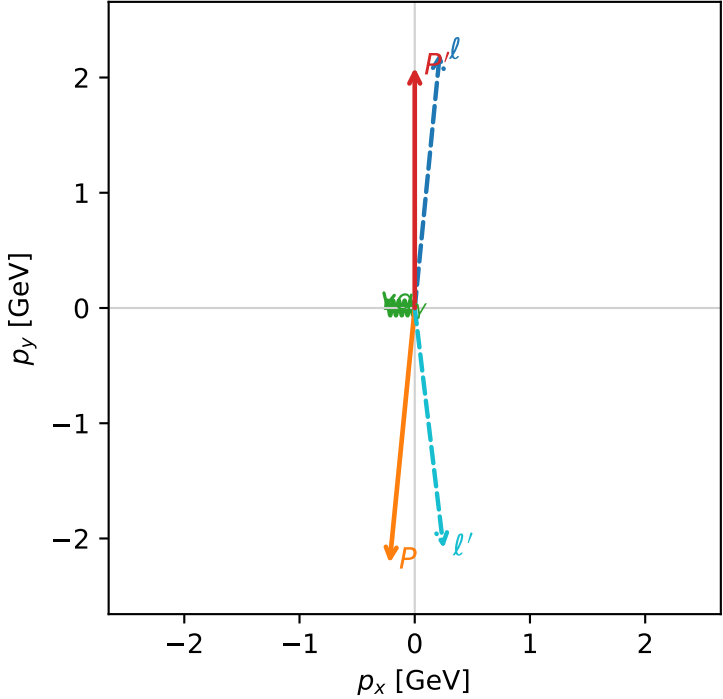


ghz_purity_low_egamma_lty_region_1 $P_{\text{GHZ}}=0.27977$
region: low_Egamma_LTy_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_{\ell_0}\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.08621$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=1.47262$, $\phi_p=4.61421$
 $E_{\gamma}=0.25$, $\phi_{\gamma}=3.14159$
 $Q^2=18.4751$, $x_B=0.941704$, $t=-18.5208$, $W^2=2.02353$, $y=0.950707$
 $\theta(\ell', \gamma)=96.8334$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.2180$ $p_y= 2.2137$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.2180$ $p_y= -2.2137$ $p_z= 0.0000$
 ℓ' $E= 2.1011$ $p_x= 0.2500$ $p_y= -2.0862$ $p_z= 0.0000$
 P' $E= 2.2874$ $p_x= 0.0000$ $p_y= 2.0862$ $p_z= 0.0000$
 q_{γ} $E= 0.2500$ $p_x= -0.2500$ $p_y= 0.0000$ $p_z= 0.0000$

Transverse plane



$h_e=+1, h_p=+1, h_{\gamma}=+1$
electron L, proton Ty

$h_e=+1, h_p=+1, h_{\gamma}=-1$
electron L, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=100.0%)

